MIL-D-17432C(SHIPS)
8 June 1965
SUPERSEDING
MIL-D-17432B(SHIPS)
17 February 1959
(See 6.3)

MILITARY SPECIFICATION

DRIVE ASSEMBLIES - ELECTROHYDRAULIC, AMMUNITION HOIST

1. SCOPE

1.1 Scope. - This specification covers electrohydraulic semi-automatic power equipment for driving vertical dredger-type ammunition hoists aboard Naval vessels.

1.2 Classification. - Hoist drive assemblies shall be of the following types, as specified (see 6. 2):

Type I - 7.5 horsepower maximum continuous rating.
Type II - 15 horsepower maximum continuous rating.
Type III -25 horsepower maximum continuous rating.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-S-901 - Shock Tests, H. L (High Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment, for Naval Shipboard Use.
MIL-C-2212 - Controllers, Alternating-Current, Naval Shipboard.
MIL-H-6083 - Hydraulic Fluid, Petroleum Base, Preservative.
MIL-P-15024 - Plates, Identification - Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manuals, Equipment and Systems.
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-M-17060 - Motors, Alternating Current, Integral Horsepower (Shipboard Use).
MIL-L-17672 - Lubricating Oil, Hydraulic and Light Turbine, Noncorrosive.
MIL-P-17869 - Pumps and Motors, Power, Oil, Hydraulic (Naval Shipboard Use).

STANDARD

MILITARY -

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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NATIONAL BUREAU OF STANDARDS

Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at Third St., New York, N. Y. 10016.)

AMERICAN GEAR MANUFACTURERS ASSOCIATION STANDARDS

(Application for copies should be addressed to the American Gear Manufacturers Association, Empire Building, Pittsburgh, Penna.)

3. REQUIREMENTS

3.1 Material. - Materials shall be suitable for the purpose intended. Materials for mechanical items sha11 be in accordance with either commercial specifications or Government specifications.

3. 1.1 If commercial specifications are used, the bill of material shall list both the commercial specification and the equivalent Government specification.

3. 1.2 Cast iron or malleable iron shall not be used for any part of the ammunition hoist drive assembly, except as approved by the bureau or agency concerned.

3.2 Design requirements. -

3.2.1 Function. - The drive assembly shall move a single or double reversible special type conveyor chain, loaded with ammunition, through a vertical, rectangular tube, the bottom end of which is shown on figures 1 and 2. Movement shall be in a hoisting or lowering direction as selected by the hoist operator. Movement of the chain shall be cyclic to permit loadlng and unloading of ammunition. This cyclic movement shall be obtained by proper arrangement of hydraulic and electrical control devices as specified in Duration of each cycle shall be 2 seconds, maximum, during which time the chain shall have moved 48 inches, but not including loading or unloading time.

3.2.2 Configuration. - The drive assembly, exclusive of external control devices, shall be suitable for mounting neatly on that part of the hoist tube shown on figures 1 and 2; to the right or left side as specified (see 6. 2).

3.2.3 Mechanical rating and efficency - Type I drive assemblies shall have a maximum continuous rating of 7.5 horsepower (hp.). Provision shall be made for mounting a 5 or 7.5 horsepower electric motor. Type II drive assemblies shall have a maximum continuous rating of 15 hp. with provision for mounting a 10 to 15 hp. electric motor. Type III drive assemblies shall have maximum continuous rating of 25 hp. The size of unit and electric motor required after determination in accordance with the method specified in table I, shall be as specified (see 6. 2). Efficiency (less electric motor efficiency) for all drive assemblies shall be a minimum of 70 percent.

3.2.4 Acceleration and deceleration control. - The control mechanism governing cyclic motion shall be designed to provide for smooth starting and stopping in both hoisting and Iowering directions. The control mechanism shall function properly for oil temperatures ranging between 30° and 160° Fahrenheit (F.). Acceleration and deceleration at full load of the drive unit in both hoisting and lowering shall be such that "shucking" of ammunition on the conveyor chain or undue shock to the hoist mechanism at any period during a cycle will not occur. Acceleration and deceleration at full load or less shall be substantially below that due to gravity, and shall be determined by test. Each power unit shall be properly adjusted at the contractor's plant.

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Table I - Method of calculating power requirements for hoist.

(a) Flight spacing	48 inches (constant)
(b) Degree of inclination from vertical	(From hoist arrangement)
(c) Length of chain	(From hoist arrangement)
(d) Number projectiles moved at full-load	(From hoist arrangement)
(e) Weight per projectile (pounds)	(From ammunition drawing)
(f) Weight of chain per 1 foot (pounds)	(From chain manufacturer)
(g) Constant speed (based on 30 cycles per minute)	(Calculated)
(h) Acceleration rate	(Assumed after test see 3.2. 4)
(i) Efficiency of drive, less electric motor	70 percent (constant)
(j) Chain weight (item (c) times item (f))	----------
(k) Weight of projectiles to be moved (cosine of item (b) times item (d) times item (e))	----------
(l) Total weight chain and projectiles (item (j) plus item (k))	----------
(m Assumed coefficient of friction	(Usually about 15 percent) except that the coefficient of friction shall be 25 percent where the trunk consists of more than one run
(n) Friction load (item (1) times item (m))	----------
(o) Required accelerating force (item (1) times item (h) divided by 32.2)	----------
(p) Total force required, accelerating (item (k) plus item (n) plus item (o))	----------
(q) Total force required, constant speed (item (k) plus item (n))	----------
(r) Input hp. required at sprocket accelerating (item (g) times item (p) divided by 550)	----------
(s) Input hp. required at sprocket, constant speed (item (g) times item (q) divided by 550)	----------
(t) Input hp. required, accelerating (item (r) divided by item (i))	----------
(u) Input hp. required, constant speed (item (s) divided by item (i))	
(v) HP rating and duty of electric motors	(Shall be a standard frame size nearest the requirement indicated by item (u), providing the motor selected is not overloaded by more than 50 percent during accelebration, item (t))

Note. -All determinations of size shall be rounded out to the nearest appropriate figure, which will permit use of type I, II or III drive assemblies.

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3.2.5 Components of drive assembly. - The drive assembly shall be so designed that it will comprise two major components as follows:

3.2.5.1 Electric motor controller. - This component shall include all devices necessary for starting, stopping, and for protecting the electric motor from overload. It shall also include any electrical devices which may be necessary, other than external devices such as limit switches, and solenoids, required, to insure functioning of the drive unit as specified herein.

3.2.5.2 Mechanical-hydraulic unit. - This component shall be a compact unit comprising devices as follows:

(a) A constant speed, continuously running electric motor.
(b) A hydraulic pump driven directly by the electric motor.
(c) A hydraulic motor capable of being driven in either direction of rotation by oil supplied from the pump through a control valve and piping, complete with lower sprocket shaft, bearings, key and reduction gearing as necessary.
(d) A control valve operating mechanism for selection of l “Hoist,” “Lower” or “Manual” clearly marked, and located so that it can be conveniently operated, This control valve may be substituted with hoist and lower push buttons in accordance with MIL-C-2212.
(e) Control solenoids.
(f) A mechanical means for emergency manual operation in the event of electric or hydraulic power failure. Where a hand crank is used, it shall be suitable for use by two men. It shall be possible to attain the maximum practicable delivery rate of the hoist with each man exerting not more than 40 pounds maximum effort on the crank. Throw of the hand crank shall be limited to 14 inches. The crank shall be detachable for stowage.
(g) A tank of sufficient capacity to contain all of the oil required in the system, plus space for oil expansion (See 3.4.10).

(h) An oil filter that will constantly filter the part of the oil being drawn into the hydraulic system.

3.3 Control- The control equipment shall provide for the following operation:

3.3.1 Starting of electric motor pump. - It shall be impossible to start the electric motor and the pump e any of the following conditions exist:

(a) While the hand crank for manual operation is engaged.
(b) While a watertight door located at the top of the hoist trunk is closed.

3.3.2 Control set for power hoisting. - It shall be impossible to hoist ammunition while any of the following conditions exist:

(a) While a projectile (or projectiles) remains on the conveyor chain at the upper terminal of the hoist.
(b) While a projectile (or projectiles) is not properly seated on the conveyor chain at the lower loading opening of the hoist (see figure 1).
(c) While safety doors, located at the lower loading opening, are open.

3.3.3 Control set for power lowering. - Power lowering of ammunition shall be initiated by simultaneous actiation of push buttons located at the top and bottom openings of the hoist. These push buttons shall include blue lens indicating lights which shall be arranged in the lowering control circuit to indicate when the opposite push button has been actuated. These indicating lights shall derive power from the ship’s 115 volt alternating current (a.c.) emergency lighting circuit. A provision shall be built in the hoist mechanism to prevent continuous lowering if both upper and lower push buttons are erroneously held in contact. It shall be impossible to initiate a lowering cycle while either of the following conditions exist:

(a) While a projectile (or projectiles) remains on the conveyor chain at the bottom opening of the hoist.
(b) While the safety doors at the bottom opening of the hoist are open.

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3.3.4 Manual hoisting.- In manual hoisting, a manual brake release or means to provide freewheeling of the hydraulic system, and visual flight position indicator showing position of the flights at any time or an indicator light in lieu of the visual flight position indicator which will be energized when a projectile has reached the upper terminal of the hoist, shall be provided. It shall be impossible to start power operation of hoist while control is set for manual operation.

3.3.5 Manual lowering. - Manual lowering of ammunition will not be required but will be acceptable where design is such that this operation is possible.

3.3.6 Pilot valves - manual operation. - Mechanical means, normally locked in the inoperative position for manual operation of any solenoid operated hydraulic pilot valve, shall be provided for emergency operation in case of casualty to the electric control or solenoid. Under this condition, the electric interlock circuits may be inoperative. The ready lights specified in 3.3.3 shall be used for signaling between the two ends of the hoist for this emergency operation. As an alternative for the emergency operation, a red light located at the lower end may be used which would be energized when a shell reaches the upper position. The red light may also be used in lieu of a visual flight position indicator.

3.4 Miscellaneous mechanical and hydraulic details. -

3.4.1 That part of the drive assembly specified in 3.2.5.2 shall be suitable for mounting at any angle from vertical to 45 degrees on the side of that part of the hoist shown on figures 1 and 2.

3.4.2 Where practicable, welded steel construction shall be used in lieu of castings.

3.4.3 The load capability of the gearing used in the drive assembly shall be computed and rated in accordance with the American Gear Manufacturers Association standards.

3.4.4 Means shall be provided to prevent the loaded conveyor chain from self-lowering when the hoist control is in neutral or in the event of rupture of hydraulic lines or power failure.

3.4.5 Main hydraulic pump and hydraulic motor shall be a positive displacement piston or vane type with constant or variable delivery. Gear pumps may be used for auxiliary service, such as control operating, auxiliary make-up, or supercharging. These hydraulic units shall be in accordance with MIL-P-17869.

3.4.6 Mounting plates of flanges shall be arranged and drilled for holding down bolts in such manner that accurate alignment with hoist drive sprockets will be insured at all times, taking into consideration shock and vibration.

3.4.7 Flexible couplings shall be provided as necessary between the electric motor and hydraulic pump and the hydraulic motor and gear reduction.

3.4.8 The chain sprocket drive shaft (see 3.2.5.2 (c)) shall be detachable from the drive assembly. That part of the shaft which fits into the drive assembly shall be splined to permit easy insertion and removal. The remainder of the shaft shall have a nominal diameter of 2 inches with a small decrease between the shoulders on which the chain drive sprockets are mounted. Chain drive sprocket detail is shown on figure 3.

3.4.9 The drive assembly case may be used for oil storage, replenishment, and expansion. The case shall be constructed of steel or other suitable materials, and shall be protected to prevent rust or corrosion and other foreign matter from collecting and mixing with the contained fluid. Drain and fill plugs shall be furnished as necessary.

3.4.10 The drive assembly case, if used for oil storage, replenishment, and expansion, shall be fitted with a suitable oil level indicator.

3.4.11 Bolted access and inspection covers shall be provided on the drive assembly case over all parts of the unit requiring inspection and maintenance. If practicable, these access and inspection covers shall be arranged to make possible their removal without draining all of the hydraulic fluid from the drive assembly case, where the drive assembly case is used for storage, replenishment and expansion of the hydraulic fluid.

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3.4.12 Drive assembly shall be designed for efficient operation (see 3.2.3) with hydraulic fluids conforming to Military symbols 2110T-H or 2135T-H of ML-L-17672.

3.4.13 Linkages, other than switch lever arms, for operating the various interlock switches and valves shal1 be furnished by the activity installing the power unit.

3.4.14 Keys, collars, wheels, nuts, bolts, studs and other parts that are liable to work loose shall be fastened in a positive manner. The use of set screws shall be avoided where possible, and where their use is unavoidable, they shall be positively secured.

3.4.15 Suitable means for lubrication of all moving parts of the drive assembly shall be provided. continuous lubrication by the circulating hydraultc fluid Is preferred.

3.4.16 Roller bearings or ball bearings shall be used throughout. The drive shaft bearing nearest the power unit shall be suitable for radial and thrust loads (see figures 1 and 2). In all applications of antifricion bearings, the assembly drawing of the power unit shall bear a certificate signed by a responsible official of the bearing manufacturing company indicating approval by his company of mounting, load and speed to which bearings will be subjected.

3.4.17 Exposed ferrous surfaces, after thorough cleaning, shall be given two coats of zinc chromate primer and one coat of enamel.

3.4.18 Force required to move any manually operated control levers shall not exceed 20 pounds, except for the emergency hand operating crank for the mechanical hydraulic unit (see 3.2.5.2 (f)).

3.4.19 Screw threads shall be class 3 fit conforming to Handbook H28.

3.4.20 Hoist projectile carriers shall be self-leveling when both hoisting and lowering with the hoist chain fully loaded, partially loaded, or empty. The self-leveling mechanism shall be adjustable to stop the carriers between the limits of 0 inch above and 1 inch below the level of the hoist tube loading sill.

3.4.21 All piping shall be designed to limit the maximum velocity of oil to not more than 25 feet per second. Bends in piping shall be to a radius not less than five times the inside diameter of the piping.

3.4.22 All piping, fittings, seals, and attachments, shall be in accordance with the best commercial practices.

3.4.23 Relief valves shall be furnished to limit the pressure in the system to not more than 10 percent above the maximum working pressure. Outside adjustments of relief valves shall be capped to prevent oil leakage and tampering. All adjustments shall be permanently made at factory or if approved durtng test on ship, all adjustment shall be set so as not to render further adjustment possible. Where secondary pressure lines are employed, relief valves conforming to the specified requirements shall also be fitted. The overflow from relief valves shall be led to the low pressure side of the systems and relief valve orifice restriction shall be provided, if such restriction is necessary in order to prevent lowering of a loaded hoist when the relief valves open.

3.4.24 Provision shall be made for attaching pressure gages for trial purposes as may be required.

3.4.25 Provlsion shall be made for eliminating alr from all parts of the hydraulic system.

3.4.26 All necessary bolts, studs, nuts, and lock washers for flanged connections shall be furnished.

3.4.27 Special care shall be taken to exclude sand, scale, and other foreign matter in order that no damage shall be done to the hydraulic pumps and motors. This applies to all fittings connected with the oil system. Any opening where foreign matter may enter during shipment or erection of the units shall be efficiently blanked before shipment

3.5 Electric details. -

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3.5.1 Electric motors. -

3.5.1.1 Motors shall conform to MIL-M-17060, and to the following:

Service	Navy A.
Horsepower	As required by driven auxiliary.
Ambient temperature	40°C.
Enclosure	Spraytight.
Speed	Constant.
Duty	As required to meet the test requirements of 4.5.
Text	Squirrel cage, design B.
Voltage	440 volts, a.c.
Number of phases	3.
Frequency	60 cycles.
Type of bearing	Ball.
Insulation	Class B or H.

3.5.1.2 Motors shall have a synchronous speed of not more than 3600 revolutions per minute (r.p.m.).

3.5.1.3 If fan-cooled motors are used, the fan shall be of metallic nonsparking material.

3. 5.2 Controllers. -

3.5.2.1 Controllers shall be suitable for the purpose intended, and shall conform to ML-C-2212 and to the following:

Service	Navy A.
Voltage	440 volts, a.c.
Ambient temperature	40°C.
Enclosure	Spraytight.
Control functions	Motor starting.
Type	Across-line.
operation	Magnetic.
Performance	Non-automatic.
Protection	Overload, and under voltage protection.

3.5.3 Master switches and interlocks. - Each controller shall be provided with the following master switches and interlocks, in accordance with MIL-C-2212:

(a) One spraytight push button - "Start", "Stop", and "Emergency Run" master switch for motor control.
(b) Interlock and auxiliary switches, of spraytight construction, one to prevent operation of motor when upper hoist watertight door is not completely open (see 3.3.1), and others as necessary to meet the specified requirements of automatic control (see 3.&2 and 3.3.3). These interlocks and auxiliary switches shall be of suitable design and either plunger- or lever-operated, as necessary to function satisfactorily with the mechanical features of the mechanism, as provided by the building yard (see figures 1 and 2).
(c) Spraytight push buttons for lowering and signaling (see 3.3.9).

3.6 Shock resistance. - Selection of materials and designs shall be such that drive assemblies will successfully withstand shock tests specified in 4.5.3.

3.7 Weight. - Special emphasis shall be placed on producing equipment under this specification which will feature minimum weight.

3.8 Identification plate. - Each drive assembly shall have attached an identification plate which shall be in accordance with MIL-P-15024, except that plastic shall not be used. The plate shall include room for a 14-digit stock number to be furnished by the bureau or agency concerned.

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3.9 Instruction charts. - An instruction chart having letters not less than 3/32 inch high shall be furnshed with each drive assembly. The chart shall describe clearly the proper procedure for filling the hydraulic system, shall give instructions as to the proper care and lubrication of the equipment, and shall be arranged for screw fastening to the lower end of the ammunition hoist trunk The chart shall be in accordance with MIL-P-15024.

3.10 Information plates. -Information plates shall be furnished for all levers on control valves to indicate function and direction of operation and at all points where hydraulic piping connections are made to properly identify the connection. These plates shall be in accordance with type B of MIL-P-15024, except that plastic shall not be used.

3.11 Repair parts. -

3.11.1 Onboard repair parts. - One set of repair parts shall be provided for one to four drive assemblies of the same type and rating, installed on each ship.

3.11.1.1 Mechanical and hydraulic. -Mechanical and hydraulic onboard repair parts sets shall be furnished as required by the contract or order and shall comprise such items as bearings, springs, gaskets, oil seals, and bushings, in number and quantities considered necessary by the contractor for maintenance of the drive assembly for a period of one year.

3.11.1.2 Electrical. - Electrical repair parts sets shall conform to MIL-C-2212 and MIL-M-17060.

3.11.2 Stock repair parts. - when specified (see 6.2), stock repair parts shall be furnished in accordace with MIL-P-15137.

3.11.3 Identification of repair parts. -

3.11.3.1 Onboard parts. - Onboard repair parts shall be clearly and permanently marked or provided with an identifying waterproof tag or label attached to the part. Small parts or other parts which would be impracticable to mark, tag, or label, may be enclosed in marked waterproof envelopes.

1.11.3.2 Stock parts. - Stock parts shall be identified in accordance with MIL-P-15137.

3.12 Drawings or sketches and design calculations shall be submitted with proposals, to furnish drive unit under this specification, whenever the design is new and/or different from previously furnished equipment. When equipment to be furnished is identical to plans already in possession of the bureau or agency concerned, all plans applicable to the equipment to be furnished, shall be fully identified. Plans of new design of equipment shall be in accordance with the following:

3.12.1 All drawings shall conform to MIL-D-963. In addition, all electrical drawings shall conform to the requirements of the applicable equipment specifications. Small detail drawings shall be grouped together on one large sheet before submission.

3.12.2 Any work done prior to complete approval of the drawings concerned will be entirely at the risk of the contractor.

3.12.3 For the purpose of scheduling fabrication and deliveries, the contractor may assume that 21 days will be required by the bureau or agency concerned for action on each drawing or groups of drawings submitted or resubmitted.

3.12.4 Drawings shall be submitted in accordance with MIL-D-963 for approval by the bureau or agency concerned. Drawings shall be submitted via the Government inspector with an accompanying letter pointing out any departures from this specification with the reason therefor.

3.12.5 A general arrangement drawing to be included in the first submission of drawings shall contain the following in addition to the assembly views of the unit:

(a) A list of the mechanical repair parts furnished.

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(b) An index of all related drawings with the contractor's number title and Bureau of Ships number.
(c) A notation of the weight of the unit and of the repair parts.
(d) A notation of duty ratings.

3.12.6 Prior to delivery of the equipment, the contractor shall submit to the Government inspector the original and one "See-Bee Process" tracing of each drawing as finally approved by the bureau or agency concerned. Such prints as may be required for installation proposes shall be provided as may be specified by the bureau or agency concerned at the time of installation. These tracings and prints shall constitute the corrected working drawings.

3.13 Technical manuals. - Technical manuals shall be furnished in accordance with type B of MIL-M-15071. A reproducible copy of the book shall be furnished.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for quality conformance. -

4.2.1 Lot definition. - For purposes of sampling, a lot shall consist of all drive assemblies of the same type offered for delivery under the contract or order.

4.2.2 Sampling for group A tests. - A sample of drive assemblies shall be selected in accordance with table II from each lot and subjected to each of the group A tests of 4.3.1 with lot acceptance based on the sampling requirements in accordance with MIL-STD-105.

Table II - Sampling for group A tests
AQL (approx.) = 1.5 percent defective.

Number of drive assemblies in inspection lot	Number of drive assemblies in sample	Number of equipments nonconforming on any group A test	
		number (defectives)	number (defectives)
15 and under	All	-	-
16 to 40	15	0	1
41 to 110	25	0	1
111 to 300	35	1	2
301 to 500	50	1	2
501 and over	75	2	3

4.2.3 Sampling for group B tests. - A sample number of drive assemblies shall be selected in accordance with table III and subjected to the group B tests of 4.3.2.

Table III - Sampling for group B tests.

Number of drive assembles	Number of drive assemblies for group B tests
110 and under	1
111 to 500	2
501 and over	3

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4.3 Quality conformance tests. -

4.3.1 Group A tests. - Each of the sample drive assemblies selected in accordance with table II shall be subjected to each of the following tests and the results of each test checked for comformance with this specification. Failure to conform to this specification for any group test shall be counted as a defect and the drive assembly shall be rejected. If the number of such nonconforming drive assemblies in any sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

Group A tests	Reference
Visual examination	4.4.1
Performance test	4.4.1
Manual operation	4.4.1
Mechanical efficiency	4.4.1

4.3.2 Group B tests. - Each of the sample drive assemblies selected in accordance with table III shall be subjected to each of the following tests, and the results of each test checked for conformance with this specification. I the event of any failure to conform to this specification for any group B test, the contractor shall correct the cause of failure on future production units and repair the deficiency in any drive assemblies not yet shipped.

Group B tests	Reference
Endurance test	4.4.2.1
Low ambient temperature test	4.4.2.2
High ambient temperature test	4.4.2.3
Shock test	4.4.3

4.4 Test procedures. -

4.4.1 Performance test. - All drive assemblies selected in accordance with table II, shall be tested for control, performance, ability to handle rated load and manual operation. Duration of this test shall be 15 minutes for each direction of operation. The test set-up may be such that the 15 minute duration in each direction may be accomplished in either of the following methods:

(a) Hoist for 15 minutes. Lower for 15 minutes.
(b) Hoist and lower alternately for a specified number of flights until 30 minutes have elapsed.

Assembly of a complete ammunition hoist is not required for these tests. However, the test shall simulate the actual installation of a drive assembly and control mechanism as regards speed and load. The load may be simulated by use of a proper weight suspended from a rope installed around a rope drum driven by the unit under test; also a flywheel, as necessary, may be used to simulate inertial loading encountered in acceleration and deceleration of the load. During this test, data shall be collected which shall be the basis for checking conformance with the mechanical efficiency requirements in 3.2.3.

4.4.2 Using the test arrangement specified in 4.4.1, drive assemblies selected in accordance with table III shall be subjected to the following tests, during which a separate electric motor of known characteristics may be employed in lieu of one of the motors intended for final installation. In these tests the drive assembly shall be operated under simulated service conditions of full load and full rate of delivery with all cams, valves, and relief valves, in such a manner that the assembly will decelerate to its minimum operating speed at each loading level and accelerate to its maximum speed during each cycle.

4.4.2.1 Endurance test. - Operate in the hoist direction under the conditions specified in 4.4.2 for 8 hours. During the first 1/2 hour of this test the temperature of the replenishment oil shall rise not more 100°F. above the starting temperature. The maximum temperature during this period shall be recorded. Operate the units in the lowering direction for 8 hours. Repeat hoisting and lowering until 40 hours hoisting and 40 hours lowering have been accomplished. For those units in which the temperature of the oil continues to rise after the first 1/2 hour of the endurance test, artificial cooling of the oil may be employed after the first 1/2 hour of the test. During this test the following readings shall be recorded at starting and at l/2-hour intervals:

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(a) Temperature of oil leaving the A-end.
(b) Temperature of oil leaving the B-end.
(c) Maximum oil pressure developed by the A-end.
(d) Speed in r.p.m. of the A-end.
(e) Speed in r.p.m. of the B-end.
(f) Power input to the electric motor.
(g) Rate of delivery in terms of pieces per minute.
(h) Noise and vibration (decibels and frequency).
(i) Rate of acceleration and deceleration (during tests specified in 4.4.2.2 and 4.4.2.3).

4.4.2.1.1 If the hydraulic drive is of the close-coupled transmission type (the pump and motor are bolted to a common end cap which is the valve manifold), the following data should be taken in lieu of the data specified in 4.4.2.1:

(a) Temperature of oil in case.
(b) Change oil pressure.
(c) Maximum developed working pressure.
(d) Input speed of electric motor.
(e) Power and current input to electric motor.
(f) Cycling time of transmission.
(g) Noise and vibration (decibels and frequency).
(h) Rate of acceleration and deceleration.

4.4.2.2 Low ambient temperature test. - Reduce the temperature of the hydraulic oil to 30° F. Operate the sample unit under condition of full load in hoist direction for at least 10 minutes. During this test the readings specified in 4.4. 2.1 shall be recorded at starting and at 3-minute intervals. Repeat this test in lowering direction.

4.4.2.3 High ambient temperature test. - Increase the temperature of the hydraulic oil to 160° F. Operate in hoist direction under full load conditions for at least 10 minutes. During this test the readings specified in 4.4.2.1 shall be recorded at starting and at 3-minutes intervals. Repeat this test in the lowering direction.

4.4.2.4 At the end of the tests specified in 4.4.2.1, 4.4.2.2, and 4.4.2.3 the sample drive assembly shall be disassembled and all parts examined for wear. Failure, defects, or unusual wear shall be reported with the test readings.

4.4.3 Shock test, - Shock tests in accordance with MIL-S-901 shall be performed on sample drive assemblies selected in accordance with table III.

4.4.4 Hydraulic pumps and motors shall be tested in accordance with MIL-P-17869 if the hydraulic pumps and motors have not been approved as integral components of an overall design of an accepted auxiliary.

4.4.4.1 Shipboard tests. - Tests shall be performed on drive assembles completely assembled to operate aboard ships. Test personnel will be furnlshed by the shipbuilder, but the drive assembly manufacturer may be required to furnish a service engineer at the discretion of the bureau or agency concerned (see 6.2).

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation and for storage of onboard repair parts.

5.1.1 Drive assemblies, electrohydraulic, ammunition hoist. -

5.1.1.1 Preservation and packing. - Preservation and packing shall be sufficient to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice.

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5.1.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier, at lowest rate, and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5.1.2 Onboard repair parts. -

5.1.2.1 Cleaning, preservation and packaging. - Repair parts shall be packaged one item per unit package unless used in sets or in quantities greater than one. Electric repair parts shall be preserved packaged as specified in 5.2.2. All packaged parts shall be identified by item, part number, quantity and stock number, if known.

5.1.2.2 Repair parts boxes. - Repair parts boxes shall conform with the requirements of type M metal boxes of MIL-B-233. The gross weight shall not exceed 200 pounds.

5.1.2.3 Index list of repair parts and tools. - An index of repair parts and tools which gives a complete itemized listing of the container content shall accompany the equipment, and shall be inserted in the interior side of the parts box cover or shall be suitably placed on the inside of the box for quick accessibility. The list shall be in a waterproof bag or shall be so treated as to be resistant to water, oil and fading.

5.2 Domestic shipment and storage or overseas shipment. -

5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment which may be required when procurement is made (see 6. 2).

5.2.1.1 Preservation and packing

5.2.1.1.1 Level A. -

5.2.1.1.1.1 Hydraulic drive unit. - Internal surfaces of the drive assembly shall be cleaned in accordance with method C-3 and exterior surfaces shall be cleaned in accordance with method C-1. The hydraulic drive unit shall be filled and internally flushed with hydraulic equipment preservative oil in accordance with MIL-O-6083, or thin film rust preventive compound conforming to P-2 of MIL-P-116, as speciified (see 6.2). The drive assembly shall be operated repeatedly for every short intervals to insure complete coating of the internal mechanisms. This preservative shall be completely drained and all openings sealed. Open flanges and other openings shall be covered with barrier materiaI and where practicable, a blank flange or plug shall be bolted or securely fastened over the barrier. Internal protection afforded by preservative (hydraulic fluid petroleum base) shall be limited to one year and drive assemblies so protected by this method shall be conspicuously labeled on the unit package and shipping container to indicate date of preservation and to warn that represervation is required at expiration of one year. Unpainted exterior surfaces shall be coated with preservative type P-1 of MIL-P-116. Exterior unpainted subassemblies having moving parts requiring protection shall be coated with preservative P-2 of MlL-P-116, and covered with barrier material.

5.2.1 .1.1.2 Component or dissembled parts. -

5.2.1.1.1.2.1 Electrical components not removed from the equipment-motors, solenoids, instruments and switches. - All openings shall be sealed with pressure-sensitive tape lAW of PPP-T-60 and all exterior surface including the tape shall be coated with electric ignition waterproofing varnish type 1AW of MIL-V-13811.

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5.2.1.1.1.2.2 Electrical components detached from basic unit. - All electrical components not attached to basic components shall be preserved and unit packaged in accordance with MIL-E-16298.

5.2.1.1.1.2.3 Mechanical components detached from basic unit. - The drive assembly shall be preserved and packaged in as near complete assembly as practicable. When partial disassembly for shipment is required, the components shall be preserved with preservative P-2 of MIL-P-116, packaged in accordance with method 1 and secured to the equipment.

5.2.1.1.2 Level C. - Equipment shall be preserved and packaged in accordance with the requirements of 5.1.1.1.

5.2.1.2 Packing (see 6.2). -

5.2.1.2.1 Level A. - The equipment, detached components and other items, packaged as specified, shall be packed in wood-cleated plywood boxes in accordance with PPP-B-601, overseas type, as nailed wood boxes in accordance with class 2 or PPP-B-621, or nailed wood crates conforming to MIL-C-104. Crates shall be lined with barrier material. Shipping containers shall be closed and strapped with flat steel strapping. The gross weight of wood boxes shall not exceed 200 pounds, unless the weight of a single item exceeds this amount.

5.2.1.2.1.1 Where a single item weight is over 200 pounds but not to exceed 500 pounds, the wood boxes shall be modified by the addition of skids and the item bolted through the skids.

5.2.1.2.1.2 Level B. - Equipment, packaged as specified shall be packed as specified in 5.2.1.2.1.1 except it shall be domestic type packing.

5.2.1.2.1.3 Level C. - Equipment, packaged as specified shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination at the lowest applicable rate. Shipping containers shall comply to the Uniform Freight Classification Rules or other regulations applicable to the mode of transportation.

5.2.1.3 Marking. - The interior packing and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.2.2 Repair parts. - Repair parts shall be preserved and packaged level A; packed level A, B or C anal marked in accordance with MIL-P-16298.

5.2.3 Service manuals. - Bulk service manuals, for shipboard electrical and mechanical equipment, shall be prepared for shipment in accordance with MIL-M-15071.

5.2.4 Drawings. - Drawings shall be prepared for shipment in accordance with MIL-D-963.)

6. NOTES

6.1 Intended use. - Electrohydraulic drive assemblies covered by this specification are intended for driving semi-automatic dredger type ammunition hoists designed for handling 3 inch or 5 inch ammunition between magazines and ready service rooms aboard Naval vessels.

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of unit required (see 1.2).
(c) Hand of unit required (determined when facing hoist loading opening) (see 3.2.2).
(d) Hoist arrangement, including degree of inclination and length of hoist in order that electric motor size may be computed from maximum load (see table I and 3.2.3).
(e) Whether stock repair parts are required (see 3.11.2 and 5.1.2.2).
(f) Drawing approval activity (see 3.12.4).
(g) Whether engineering services are required (see 4.4.4.1).
(h) Selection of applicable level of preservation, packaging and packing required (see 5.1, 5.2.)

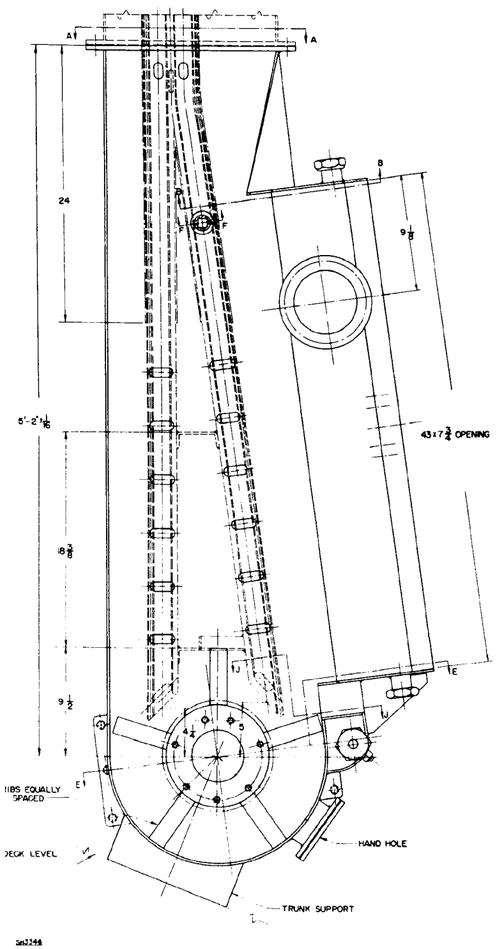
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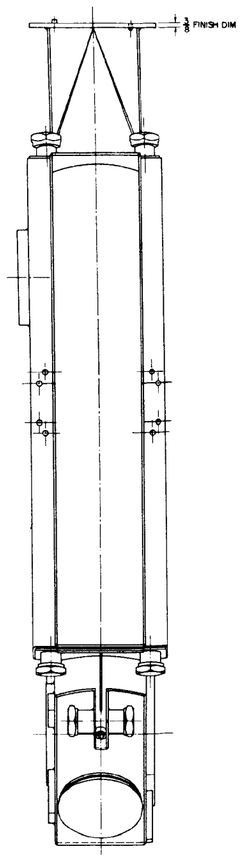
(i) Type preservative to be used for drive unit (see 5.1.1.2).
(j) Level of packing for bulk quantities of technical manual (see 5.2.3).
(k) Level of packaging and packing for drawings (see 5.2.4).

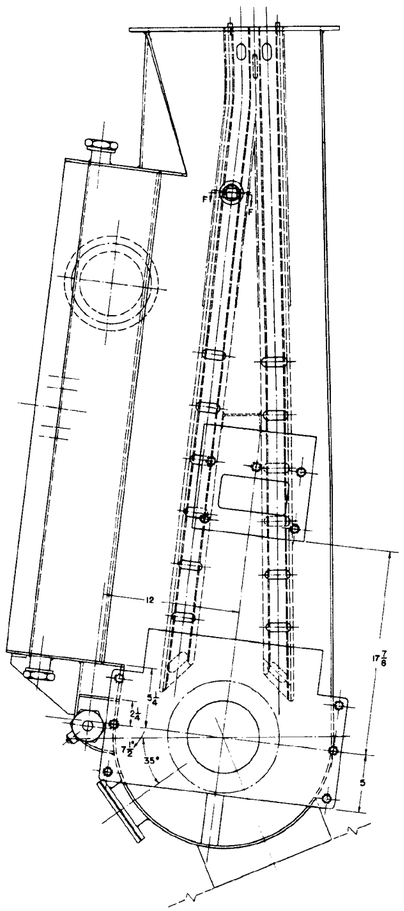
6.9 CHANGES FROM PREVIOUS ISSUE. - THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

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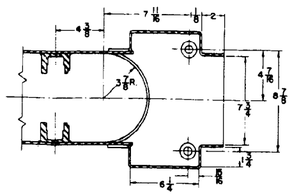




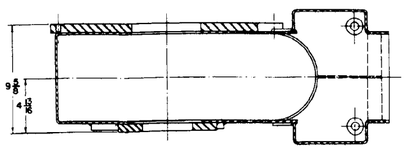


STD AMM HOIST (SINGLE)
FOR 3"-50 & 3"-70 CAL TANKS
& 3"-70 CAL BARE CARTRIDGE
TRUNK & TRACKS - LOWER END

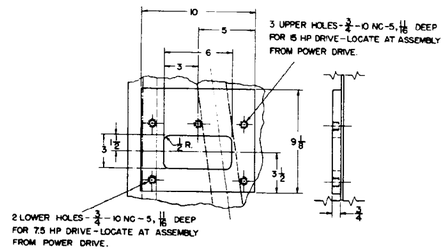
Figure -1



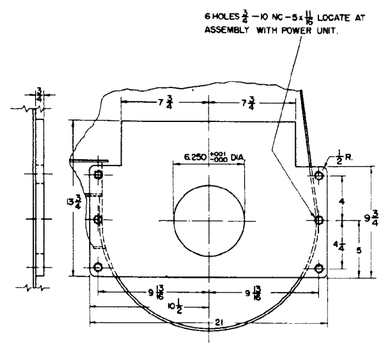
SECTION "B-B"



SECTION "E-E"

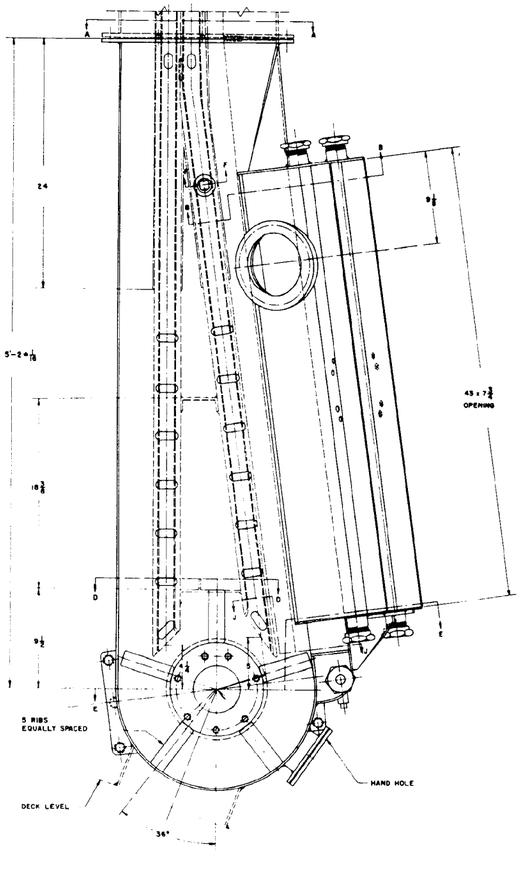


DOUBLER PLATE - FOR DRIVE & CONTOL UNIT

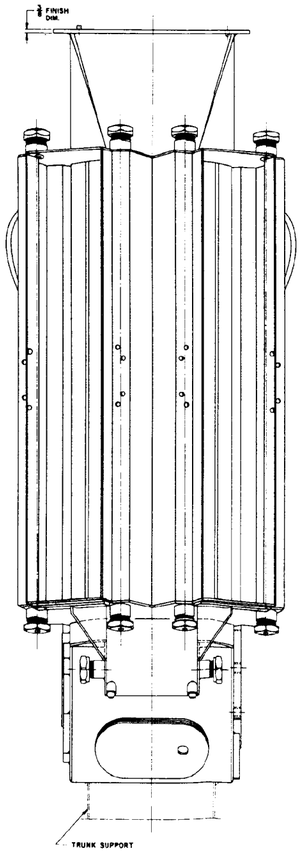


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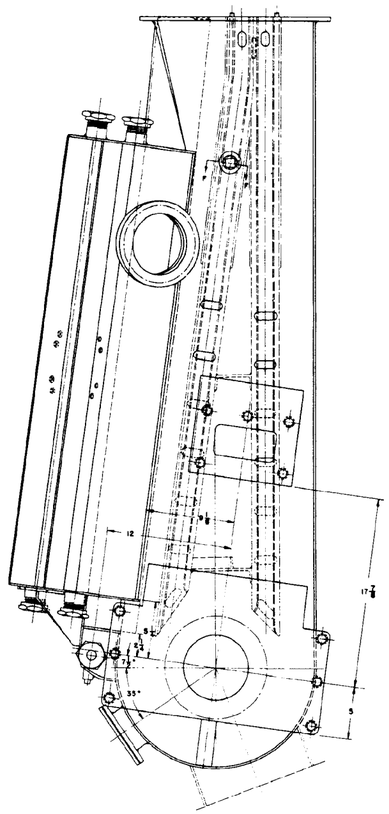
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LEFT SIDE ELEVATION



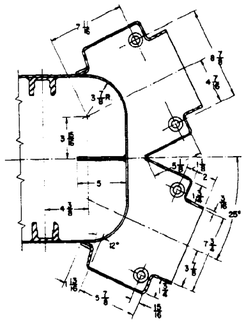
FRONT ELEVATION



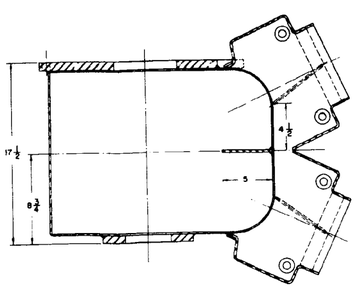
RIGHT SIDE ELEVATION

STD AMM HOIST (DOUBLE) RH DRIVE
FOR 3"-50 & 3"-70 CAL TANKS &
3"-70 CAL BARE CARTRIDGES
TRUNK & TRACKS - LOWER END

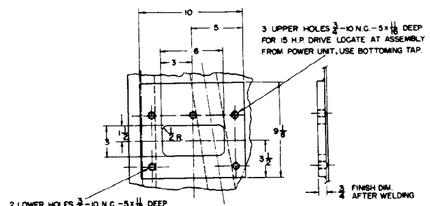
Figure – 2



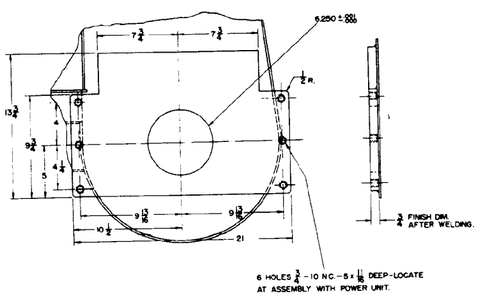
SECTION "B-B"



SECTION "E-E"



DOUBLER-FOR DRIVE & CONTROL UNIT



DOUBLER-FOR DRIVE & CONTROL UNIT

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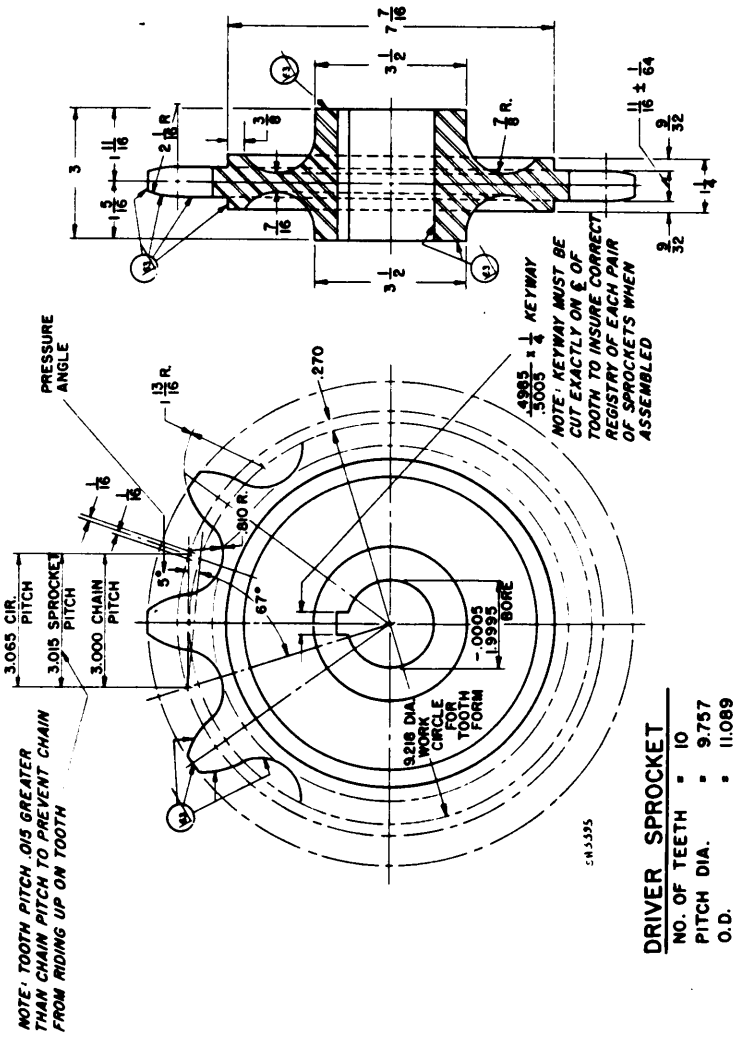


Figure – 3

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

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